Disease Ontology Expansion: Honoring Contributors, Updating Data & Resources and Strengthening FAIR Data

Claudia M. Sánchez-Beato Johnson, J. Allen Baron, Dustin Olley, Mike Schor, Lynn M. Schriml

University of Maryland School of Medicine, Institute for Genome Sciences, Baltimore, USA.

Engage with the DO

Ask us questions!!

Discuss Ideas!!

Suggest updates!!

DISEASE ONTOLOGY KNOWLEDGEBASE

GLOBAL CORE BIODATA RESOURCE

INTRODUCTION

The Disease Ontology Knowledgebase (DO-KB, https://www.disease-ontology.org/) is a Global Core Biodata Resource that facilitates disease data integration, standardization, and analysis across the biomedical research ecosystem. The Disease Ontology (DO) provides a stable, etiology-based framework that unifies data on disease nomenclature, classification, and descriptive features. Here, we present updates to the DO-KB, focusing on internal content improvements, recognition of key contributors, and enhancements that support resource users while strengthening the FAIR principles (Findability, Accessibility, Interoperability, and Reusability). These updates further enrich the interconnected web of data contributors, resource users, and integrated biomedical resources, reinforcing the role of the DO-KB as a leader in advancing disease knowledge.

Programmatic Data Retrieval

DO-KB Tools NEW! API

Submit Complex Searches: mine semantically defined disease-to-disease connections via annotated features

Extract DO datasets

Search label, ID, xref.

Generate ML-ready Datasets.

-RESTful interface
-Mine relationships
-IDs, Labels
-Imports, Prefixes

Uses the same search as DO-KB's faceted search, when searching imports.

Servers
https://api.disease-ontology.org/v1

Swagger Supported by SMARTBEAR

Home > DO-KB Home > API Documentation

The Disease Ontology API

The Disease Ontology API provides a powerful RESTful interface typically used programmatically to interact with disease-related data. This web interface, powered by SmartBear's SwaggerUI, allows you to explore and understand the API's functionality without writing code. Our API specification is OpenAPI 3.1 compliant, ensuring a modern and standardized approach to interacting with the data. Through this platform, you can easily perform operations such as retrieving specific disease terms, conducting searches similar to the site's advanced search page, and utilizing faceted search to filter results by different categories. Additionally, a range of other operations are available, giving you full access to the rich dataset of the Disease Ontology for your exploration and learning.

Our API can be invoked programmatically by pointing your client to **https://api.disease-ontology.org/v1** and requesting an endpoint as defined below.

Swagger

Disease Ontology - API 1.0.0 OAS 3.1
/static/js/do-api-spec.json

The Human Disease Ontology API Specification.

- Learn more about the Human Disease Ontology (DO)

Contact Us - Website
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Servers
https://api.disease-ontology.org/v1

info
GET /info Get DO API Metadata

terms
GET /terms Get all disease terms
GET /terms/{termId} Find Term by ID

DO Additions and Augmentations

The DO continually adds new content (disease terms, definitions, xrefs), axioms (via imports).

DO curation includes nomenclature updates and integration of disease re-classifications.
New in 2025: (1) utilizing 'en' and 'es' languages tags; (2) Spanish DO; newly available: **doid-international.owl & doid-es.owl** (3) annotating Acronyms; (4) revising OMIM --> MIM prefix; (5) **recent targeted curation** (e.g., OMIM phenotypic series additions, updates; glycosylation-related diseases; viral infectious diseases, updated DO DISDRIV and (6) new childhood cancers **DO_childhood_cancer_slim**.

Early-March 2025 release: This release of the Human Disease Ontology includes 11,837 disease classes, 9,519 with textual definitions (80.4%). See details in DO's monthly release notes: https://github.com/DiseaseOntology/HumanDiseaseOntology/releases

ACRONYMS

Synonyms -- that are acronyms -- newly annotated in DO

label
retinitis pigmentosa 14

id
DOID:0110381

has_obo_namespace
disease_ontology

definition
A retinitis pigmentosa that has_material_basis_in mutation in the TULP1 gene on chromosome 6p21.
dctype
'curator inference from journal publication'
database_cross_reference
url:https://www.ncbi.nlm.nih.gov/pubmed/9462751

has_exact_synonym
RP14
has_synonym_type
acronym

LANGUAGE TAGS

TAGS: EN (ENGLISH)/ ES (ESPAÑOL/SPANISH)

Annotations

label [language: en]
physical disorder

label [language: es]
trastorno físico

English and Spanish Labels

id
DOID:0080015

has_obo_namespace
disease_ontology

definition [language: en]
A disease that has_material_basis_in a genetic abnormality, error with embryonic development, infection or compromised intrauterine environment.
dctype
'curator inference from Wikipedia'
database_cross_reference
url:http://en.wikipedia.org/wiki/Congenital_disorder

Definitions

definition [language: es]
Una enfermedad que tiene_su_base_material_en una anomalía genética, un error con el desarrollo embrionario, una infección o un entorno intrauterino comprometido.

has exact synonym [language: en]
congenital disorder

has exact synonym [language: es]
trastorno congénito

Synonyms

https://disease-ontology.org/community/

Home > Community > Registry of Contributors

Registry of Contributors

The contributions of scientific researchers, clinical practitioners, and organizations are invaluable to the development and maintenance of the Human Disease Ontology. The collective knowledge, perspectives, and expertise from this broad community ensure the ontology remains comprehensive, accurate, up-to-date, and useful as it is used to interconnect disease research around the globe.

🎉 As a community we celebrate and express gratitude! 🎉

Resources contributing terms to the DO.
Dates reflect the earliest collaborative efforts.

Recognizing the many individual contributors, through GitHub Issues, collaborations, UMSOM Medical students, DO alumni, DO team members.

Individual Contributors

Show 10 entries — Search:

Name	Links	Affiliation
Andra Waagmeester	GitHub: andrawaag ORCID: 0000-0001-9773-4008	Amsterdam University Medical Center
Anna V. Anagnostopoulos	GitHub: anna-anagnostop ORCID: 0000-0002-6490-7723	Mouse Genome Informatics
Anu Pujar		
Asiyah Yu Lin	GitHub: linikujp ORCID: 0000-0003-2620-0345	Axle Research and Technology
Becky Jackson		Institute for Genome Sciences, University of Maryland School of Medicine
Brooke Hyman		University of Maryland School of Medicine
Carol Greene		University of Maryland School of Medicine
Carrie McCracken		Institute for Genome Sciences, University of Maryland School of Medicine
Charles Tapley Hoyt	GitHub: cthoyt ORCID: 0000-0003-4423-4370	Northeastern University
Chris Le		
Name	**Links**	**Affiliation**

Showing 1 to 10 of 52 entries — Previous 1 2 3 4 5 6 Next

Resource & Group Contributors

Show 25 entries — Search:

Name	Link	Contributing Since
Alliance of Genome Resources	ALLIANCE of GENOME RESOURCES	2016
Cancer Knowledgebase	The Jackson Laboratory	2014
Clinical Interpretation of Variants in Cancer	CIVIC	2015
Common Fund Data Ecosystem	CFDE	2018
FlyBase	FlyBase	2014
HIVE Lab (Mazumder Research Group)	HIVE	2014
Immune Epitope Database	IMMUNE EPITOPE DATABASE AND ANALYSIS RESOURCE	2009
Mondo Disease Ontology	Mondo	2017
Mouse Genome Informatics	MGI	2014
Neuroscience Information Framework	NIF	2011
Online Mendelian Inheritance in Man	OMIM	2011
Protein Ontology	PRO Protein Ontology	2013
Rat Genome Database	RGD	2006
Reactome	reactome	2008
Saccharomyces Genome Database	Saccharomyces GENOME DATABASE	2017
Wikidata	WIKIDATA	2014
Wormbase	WormBase	2011
Xenbase	Xenbase	2019
ZFIN	ZFIN	2011
Name	**Link**	**Contributing Since**

Showing 1 to 19 of 19 entries — Previous 1 Next

Last updated: March 2025

Critical analysis of disease nosology in the DO

https://www.disease-ontology.org/resources/nosology/

Disease Ontology (DO) Nosology Program

The Nosology training program is coordinated by the Human Disease Ontology (DO) Knowledgebase project (https://disease-ontology.org/), hosted at the University of Maryland School of Medicine, Institute for Genome Sciences, led by the DO's Clinical Leaders (Dr. Carol Greene (Pediatrics), Dr. Rima Koka (Pathology), Dr. Richard Lichenstein (Emergency Medicine, Pediatrics), Katharine Bisordi, MS, MGC (Genetic Counselor, Obstetrics, Gynecology and Reproductive Sciences)) and DO's PI (Lynn M. Schriml, Ph.D.) at the Institute for Genome Sciences.

The Nosology training program will utilize the Human Disease Ontology as a resource for structuring the review and highlighting the complexity of disease nosology. One of the advantages of the DO's structured knowledge is that it can be utilized as a teaching tool.

How are diseases named ?

How to organize disease knowledge ?

DO Nosology Program Objectives

A mentored, multidisciplinary, educational project designed to:

- advance cross-disciplinary nosology knowledge
- gain new perspectives on the classification of complex diseases
- enhance trainee and mentor appreciation and understanding of the nuances and complexities of disease nosology
- improve entries (disease descriptions) in the Disease Ontology as the result of clinically-led, structured review
- utilize the DO as a resource for structuring the review and highlighting the complexity of disease nosology

Continuing Medical Education

Nosology Team Review Activities

- Team: 3-4 Mentees and their Mentor
- To conduct ~ 2 hours/month for nosology reviews over one semester
- The teams will meet monthly for up to three months, to:
  (1) select a disease to review
  (2) review the current DO classification
  (3) discuss nosology challenges
  (4) provide their classification recommendations, to the DO Clinician Team Leader for feedback
  (5) Team leader (Carol, Rima, Rich, Katie) - submits review outcome to the DO

COMMUNITY

The Disease Ontology values your feedback!
Take a few moments to complete our:

<----2025 Annual DO-KB User Survey

We want to know what disease information matters most to you. Your input will help shape our data, tools, and future updates.

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